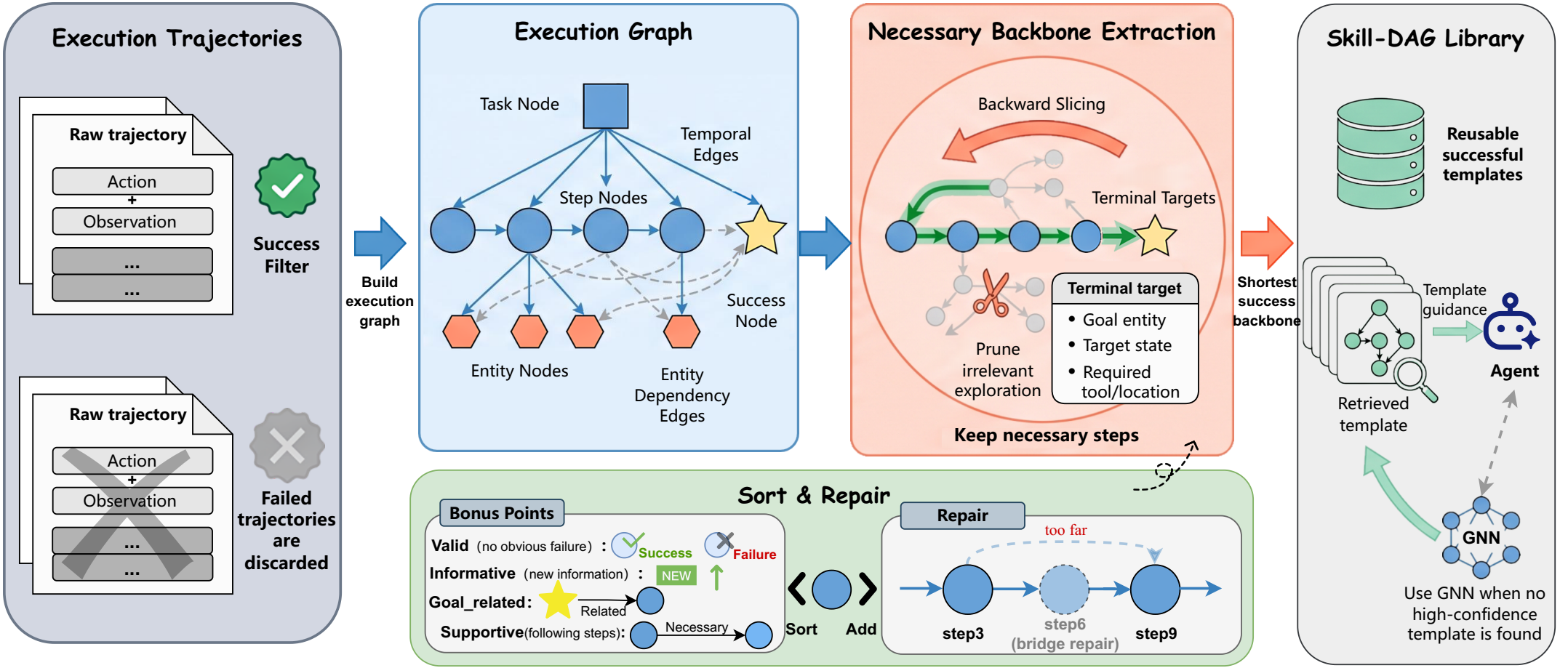

Execution Trajectories
Raw trajectory
Action
+
Observation
...
...
Success Filter
Raw trajectory
Action
+
Observation
...
...
Failed trajectories are discarded
Build execution graph
Execution Graph
Task Node
Temporal Edges
Step Nodes
Success Node
Entity Nodes
Entity Dependency Edges
Necessary Backbone Extraction
Backward Slicing
Terminal Targets
Prune irrelevant exploration
Terminal target
• Goal entity
• Target state
• Required tool/location
Keep necessary steps
Shortest success backbone
Skill-DAG Library
Reusable successful templates
Template guidance
Agent
Retrieved template
GNN
Use GNN when no high-confidence template is found
Sort & Repair
Bonus Points
Valid (no obvious failure) :
Success
Failure
Informative (new information) :
NEW
Goal_related:
Related
Supportive(following steps):
Necessary
Sort
Add
Repair
too far
step3
step6
(bridge repair)
step9